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by Mohinikanta Sahoo

A Modular Framework for Prompt Injection Detection in GPT-Based AI Systems

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Abstract. Large Language Models are now everywhere. They drive chatbots, coding support, self-autonomous workflows, and other platforms that assist large businesses. Although they are impressive, they do have a big flaw – prompt injection attacks. In other words, someone can insert malicious code and manipulate the model to bypass its safety parameters, disclose private information or generate inappropriate content.

The majority of defenses that are proposed in the literature are based on either simple machine learning techniques or heavy transformer models. The problem is that the lighter ones are not sufficiently powerful, and the heavier ones take too long and consume a lot of computing power to be feasible.

In this research, a hybrid prompt injection detection system is proposed using ensemble learning and smart semantic analysis. Uses TF-IDF and three common classifiers: Logistic Regression, SVM, and Stochastic Gradient Descent to quickly detect the obvious threats. Next, a finely-tuned XLM-RoBERTa transformer scrutinizes deeper, revealing hidden or tricky malicious prompts. They collaborate by combining their results using a weighted scoring and a threshold-based engine which categorises each prompt as ALLOW, REVIEW or BLOCK.

This system has been evaluated on internal and external datasets and has shown high precision, high accuracy, high recall and a good F1-score in the case of subtle semantic attacks. It is scalable, modular, and ready to deploy, making it easier to protect GPT-powered systems from prompt injection attacks.

Keywords: Prompt Injection Detection, Large Language Models, Ensemble Learning, XLM-RoBERTa, AI Security

1 Introduction

Today, Large Language Models (LLMs) are in use all around us, from hospitals to business applications and from customer support to coding tools, to classrooms. They've indeed made things smoother and far more effective in a relatively short period of time. However, with all the added convenience comes a big load of security issues.

Lately, prompt injection attacks are making a lot of problems. Somebody sneaks in a witty suggestion that makes the model forget to follow the safety rules, bypass security checks, or divulge information that it shouldn't. The tricky part? These types of models are not very good at distinguishing between a legitimate command and just some random man who's just trying to cause trouble. Prompt injection problems keep getting worse as AI tools connect with additional software, APIs, and databases.

What used to be considered "old school" security tactics—such as keyword filtering or fast rule checking—just don't work anymore. Attackers continue to devise new methods to bypass them, encoding new instructions, bending language etc. or inserting things underneath filters. Even common machine learning tools may fail to see the finer details, and transformer-based only models are not fast enough to provide protection in real time.

So, this is where this research comes in. It proposes a novel layered system: Hybrid detection system. It essentially combines the traditional machine learning models with powerful transformers for semantic analysis. With the help of a specialized XLM-RoBERTa model, this framework achieves even greater performance in identifying sneaky prompts, outperforming the classic algorithms and TF-IDF feature extraction.

1.1 Motivations

Large Language Fashion (LLMs) took off in all sorts of applications, yet it additionally opened the door to new security threats—especially injection attacks on altogether systems based specifically on GPT Display of sentences. Additionally, security tools for AI systems in the real world should be robust, flexible, and resistant to both obvious and subtle intrusions, minimizing false alarms.

1.2 Objectives

This research aims at achieving the following:

1. Design a lightweight prompt injection detection system based on TF-IDF and machine learning techniques.

2. To enhance the detection performance by ensemble learning techniques.

3. Add semantic understanding to prompts using transformer.

4. To create a hybrid scoring system that incorporates machine learning and transformer predictions.

5. To minimize false negatives and enhance robustness towards obfuscated attack.

6. To measure the framework with internal and external data sets.

7. To design a modular structure, which can be applied in practice.

1.3 Original Contributions

The significant achievements of this work are that:

- Building a lightweight TF-IDF based prompt injection detection model.
- Building an ensemble of Logistic Regression Classifier, SVM Classifier, and SGD Classifier.
- Finally, the fine-tuning of XLM-RoBERTa for semantic prompt injection detection.
- Designing a hybrid weighted score fusion of ensemble and transformer outputs.
- The introduction of a threshold-based decision engine (ALLOW, REVIEW, BLOCK).
- Testing of the framework using external data set to assess robustness and generalization.
- Crafting a modular and deployable security solution for AI systems that use the GPT architecture.

1.4 Paper Layout

The rest of the paper is organized as follows:

In section 2, related work and existing prompt injection defense techniques are discussed. Section 3 is an explanation of the proposed framework, dataset details and methodologies employed in the system. The experimental setup, evaluation metrics and performance analysis are presented in Section 4. Lastly, Section 5 wraps up the paper and gives some scope and future improvement ideas.

2 Literature Survey

Prompt injection attacks have become one of the major security concerns in Large Language Model (LLM) based systems. The attacks trick the user prompts into circumventing the instructions the system gave, or extracting confidential information or triggering unanticipated actions from AI models. With the increasing use of GPT-based systems in the real world, researchers have begun to investigate how to effectively protect against these attacks.

Maloyan et al. conducted a study on vulnerabilities of agentic AI systems and demonstrated attacks on tools, workflows, and external integrations used in conjunction with LLMs using a technique known as prompt injection. They showed that it is very challenging for models to distinguish trusted instructions from untrusted user input.

McHugh et al. discussed the types of hybrid AI threats and limitations of conventional defense strategies against sophisticated prompt injection techniques. They highlighted the need for multi-layered security strategies to safeguard LLM systems.

To protect prompt injection attacks in LLM-based applications, Lan et al. proposed a prompt injection detection framework that applies machine learning methods and semantic analysis. They showed that the use of more than one detection signal increases the detection capability and decreases unsafe outputs.

Rahman et al. used multilingual transformer embeddings with machine learning classifiers for malicious prompt detection. Their findings indicated that they achieved better contextual understanding than with traditional standalone models.

To mitigate over-defensive behavior in prompt injection detection systems, while ensuring high security performance, Li et al. presented a framework called InjecGuard.

While some existing methods yield success, others are limited to simple machine learning tools or adopt the heavy-weight transformer architectures. Furthermore, a few systems are not able to handle obfuscated and context-aware attacks. The proposed approach aims to overcome these limitations by incorporating ensemble learning and transformer-based semantic analysis to enhance the robustness, contextual understanding, and applicability of real-world deployments.

3 Proposed System/Model

The proposed model is a multi-layer prompt injection detection system, employing light machine learning techniques along with a transformer-based semantic analysis. The framework filters out user prompts before they pass through to the target LLM and categorizes them as safe or malicious via hybrid scoring and threshold-based decision making.

3.1 Dataset Description

In this study, we have used both manually collected and publicly available prompt injection datasets to generate the dataset. First, a custom set of 1262 prompts (both harmless and malicious) were created. In addition, 53,616 prompts were collected from Hugging Face datasets and used for training the ensemble learning models. The framework's ability to generalize was examined using a separate testing set with 10,775 prompts from Hugging Face for external evaluation.

If a transformer-based training, the custom dataset and Hugging Face datasets were merged and preprocessed to eliminate unnecessary formatting inconsistencies and special characters and emojis. We preprocessed and balanced the dataset and selected 6,256 samples to train the XLM-RoBERTa transformer model. The transformer model was then tested on the same external test set consisting of 10,775 prompts.

3.2 Schematic Layout of the proposed system/model

The proposed framework follows a layered architecture consisting of preprocessing, ensemble learning, transformer-based semantic analysis, and threshold-based decision making.

User Prompt
↓
Preprocessing Layer
↓
TF-IDF Vectorization
↓
Ensemble Learning Layer
(LR + SVM + SGD)

↓
Transformer Semantic Layer
(XLM-RoBERTa)
↓
Hybrid Weighted Scoring
↓
Decision Engine
(ALLOW / REVIEW / BLOCK)

3.3 Methodologies Used

This project is based on the hybrid NLP based approach for the detection of prompt injection in the GPT based AI systems. The methodology consists of text preprocessing (cleaning, normalization, tokenization, and noise removal), feature extraction using TF-IDF, ensemble machine learning classification using logistic regression, SVM, and SGD models, and semantic analysis using the XLM-ROBERTa transformer model. A weighing hybrid scoring method to combine machine learning and transformer outputs to produce a final risk score. Prompts are categorized as Allow, Review, or Block based on predefined thresholds. Accuracy, Precision, Recall, F1-score and Confusion Matrix are used for evaluating the system performance.

4 Experimentation and Model Evaluation

This proposed framework was tested with internal and external datasets of benign and malicious prompts. Each ensemble model and final hybrid model integrating transformer-based semantic analysis was analyzed.

4.1 Experimental Setup

The framework was built with Python, Scikit-learn, PyTorch and Hugging Face Transformers. The ensemble model has been trained with 53616 samples from Hugging Face, in addition to 1262 samples that were manually collected. The combined dataset was preprocessed to remove unnecessary formatting, special characters, and emojis, and from which 6256 samples were selected for training in the transformer training.

The features were extracted using TF-IDF vectorization and XLM-RoBERTa was fine-tuned for semantic prompt injection detection in the ensemble layer. Ensemble and transformer outputs were used in final hybrid framework with weighted scoring. The external evaluation was done with a separate test set of 10,775 prompts.

4.2 Results and Outputs

The ensemble model obtained external accuracy: 97.21%, precision: 97.95%, recall: 96.89% and F1-score: 97.42%. The final hybrid system obtained the external accuracy of 96.93%, precision of 96.69%, recall of 97.70% and F1-score of 97.19%.

The hybrid approach demonstrated better semantic understanding and enhanced capabilities to detect obfuscated prompts with an injection attack. The confusion matrix analysis results also indicated that the false negative was reduced as compared to the ensemble-only model.

4.3 Validation/System Performance Evaluation

The framework was tested in terms of accuracy, precision, recall, f1 score and confusion matrix analysis. External dataset testing was conducted to assess the general ability of the system to generate responses to test prompts that are not included in the training set.

The hybrid framework showed improved resilience to both the context-aware and semantically manipulated attacks by minimizing the number of undetected malicious prompts. The classification of prompts as ALLOW, REVIEW, and BLOCK decisions also occurred using threshold based classification.

4.4 Discussions on Contributions

The proposed framework is based on light ensemble learning and transformer-based semantic analysis to enhance the prompt injection detection. The ensemble layer is responsible for rapid and effective detection and the transformer layer is used to enhance the understanding of the malicious prompts in context.

The hybrid architecture provides practical deployment capability, reduces false negative and is robust against obfuscated attacks. Furthermore, its flexibility and modularity make it easy to integrate into AI systems based on GPT models.

5 Conclusion and Future Scope

The proposed framework successfully showed that prompt injection attacks could be detected through a lightweight machine learning in combination with the use of transformer-based semantic analysis. The experimental results demonstrated that the TF-IDF and ensemble learning models are effective and rapid in detecting attacks and the transformer layer is effective in understanding the semantics and minimizing false negative in complex and obfuscated attacks. The hybrid method showed high detection performance, robustness and practicability in implementation.

Future Scope

- Increasing sample size of prompt injection samples across a range of diverse and adversarial examples.
- Advanced robustness testing: Red team/adversarial evaluation techniques.
- Investigating light cascaded architectures for quicker real-time deployment.
- Creating low latency APIs or plugins for integration into the production AI system.
- The addition of explainability features for interpretability of detection decisions.
- Continuing the development of the framework of multilingual prompt injection detection.
- Developing ongoing learning mechanisms to respond to changing attack methods.

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